ANALYSIS I — SHEET 5 (for Week 7 tutorials)
Subsequences, Cauchy Sequences

1. Let

$$s_n = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots + \frac{(-1)^{n+1}}{2n-1}.$$

Show that (s_n) is Cauchy and hence convergent to a limit L. Show that $2/3 < L < 13/15$.

2. Let (x_n) and (y_n) be real Cauchy sequences and $c \in \mathbb{R}$. Without making reference to the Cauchy convergence criterion, show that the sequences

$$(x_n + y_n), \qquad (cx_n), \qquad (x_n y_n),$$

are Cauchy. Show further that $(1/y_n)$ is Cauchy provided (y_n) does not converge to 0. [*Hint: show that a tail of* (y_n) *is bounded away from* 0.]

3. For $n \geqslant 1$, let

$$a_n = \int_1^n \frac{\cos x}{x^2}\, \mathrm{d}x.$$

Prove, for $m \geqslant n \geqslant 1$, that $|a_m - a_n| \leqslant n^{-1}$ and deduce that (a_n) converges. By integration by parts, or otherwise, demonstrate the existence of

$$\lim_{n\to\infty} \int_1^n \frac{\sin x}{x}\, \mathrm{d}x.$$

4. (i) Let (a_n) be a sequence such that the subsequences (a_{2n}) and (a_{2n+1}) both converge to $L \in \mathbb{R}$. Show that (a_n) also converges to L.

(ii) Let (b_n) be a sequence such that each of the subsequences $(b_{2n}), (b_{2n+1}), (b_{3n})$ converges. Need (b_n) converge? Either provide a proof or a counter-example.

(iii) Let (c_n) be a sequence such that the subsequence (c_{kn}) converges for each $k = 2, 3, 4, \ldots$ Need (c_n) converge? Either provide a proof or a counter-example.

(iv) Let (d_n) be a sequence with uncountably many convergent subsequences. Need (d_n) converge? Either provide a proof or a counter-example.

5. (i) Sketch the curve $y = x^5 - 6x + 2$ and using the sketch explain why the equation $x^5 - 6x + 2 = 0$ has three real roots $\alpha < 0 < \beta < 1 < \gamma$. [You cannot yet prove this rigorously; what facts do you think you would need to know about the function $x \mapsto x^5 - 6x + 2$ to be able to treat this rigorously?]

Let $x_1 = c$ and define recursively $x_{n+1} = \left((x_n)^5 + 2\right)/6$.

(ii) Show that *if* the sequence (x_n) converges to a limit L then $L^5 - 6L + 2 = 0$.

(iii) Sketch the graphs $y = x$ and $y = (x^5 + 2)/6$ on the same axes. Using the graphs say whether, and to what, (x_n) converges in terms of the initial value c. [Again this part need not be rigorous.]

(iv) For the remainder of the question we will set $c = 0$. Show that $0 \leqslant x_n < 1$ for all n, and that

$$|x_{n+1} - x_n| \leqslant \frac{5}{6}\,|x_n - x_{n-1}| \leqslant \cdots \leqslant \frac{1}{3}\left(\frac{5}{6}\right)^{n-1}.$$

[*Hint:* $a^k - b^k = (a-b)\left(a^{k-1} + a^{k-2}b + \cdots + +ab^{k-2} + b^{k-1}\right)$.]

(v) Let $n > m \geqslant 1$. Use the triangle inequality to show that

$$|x_n - x_m| \leqslant \frac{1}{3}\left[\left(\frac{5}{6}\right)^{n-2} + \left(\frac{5}{6}\right)^{n-3} + \cdots + \left(\frac{5}{6}\right)^{m-1}\right] \leqslant 2\left(\frac{5}{6}\right)^{m-1}.$$

(vi) Deduce that (x_n) is Cauchy and that $x_n \to \beta$.

Extension questions (optional)

6. Find a reduction formula for the integrals $I_n = \int_0^1 \frac{x^n}{1+x^2}\,\mathrm{d}x$ and deduce that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots + \frac{(-1)^n}{2n+1} + \cdots .$$

(This result was first shown by the Indian mathematician Madhava (c.1340-1425). In the west, three centuries later, the series was independently found in 1671 by the Scottish mathematician James Gregory and in 1674 by Leibniz.)

7. For $n \geqslant 1$ let k, m be the natural numbers such that $n = 2^{k-1}(2m-1)$, and define $a_n = k/(m+k)$.

(i) Show that for every $y \in \mathbb{Q} \cap (0,1)$, and every $N \in \mathbb{N}$, there exists $n \geqslant N$ such that $a_n = y$.
(ii) Show that for every $x \in [0,1]$, the sequence (a_n) has a subsequence which converges to x.

8. Give examples of double sequences $(a_{m,n})$ with the following properties:
(i) $\lim_{m\to\infty}(\lim_{n\to\infty} a_{m,n})$ and $\lim_{n\to\infty}(\lim_{m\to\infty} a_{m,n})$ both exist and are unequal.
(ii) $\lim_{m\to\infty}(\lim_{n\to\infty} a_{m,n})$ exists and the sequence $b_n = \lim_{m\to\infty} a_{m,n}$ is well-defined and $b_n \to \infty$.
(iii) $\lim_{m\to\infty}(\lim_{n\to\infty} a_{m,n})$ exists and the sequence $b_n = \lim_{m\to\infty} a_{m,n}$ is bounded but divergent.
(iv) $\lim\limits_{m,n\to\infty} a_{m,n}$ exists but $\lim_{m\to\infty}(\lim_{n\to\infty} a_{m,n})$ doesn't.

9. Let (b_m) be a real sequence and $(a_{m,n})$ be a double sequence of real numbers. Suppose that

- for all $m \in \mathbb{N}$, $a_{m,n} \to b_m$ as $n \to \infty$.
- $|a_{m,n} - a_{m,r}| < 2^{-n}$ whenever $0 \leqslant n \leqslant r$.
- for all $n \in \mathbb{N}$, $\lim_{m\to\infty} a_{m,n}$ exists.

(i) Show that $|a_{m,n} - b_m| \leqslant 2^{-n}$ for all $m, n \in \mathbb{N}$.
(ii) Deduce that $\lim_{m\to\infty} b_m$ exists.

10. Let $B[0,1]$ denote the set of bounded real functions $f\colon [0,1] \to \mathbb{R}$. We define $d\colon B[0,1]^2 \to \mathbb{R}$ by

$$d(f,g) = \sup_{0\leqslant x\leqslant 1} |f(x) - g(x)| \qquad \text{and we write} \qquad \|f\| = d(0,f).$$

(i) Show that $B[0,1]$ is an algebra – that is $B[0,1]$ is a real vector space which is also closed under multiplication.
(ii) Show that d is a well-defined metric on $B[0,1]$.
(iii) Show that $\|fg\| \leqslant \|f\| \times \|g\|$.
(iv) Let (f_n) be a Cauchy sequence in $B[0,1]$; that is $d(f_n, f_m) \to 0$ as $m, n \to \infty$. Show that $(f_n(0))$ is a real Cauchy sequence.
(v) Carefully deduce that a Cauchy sequence of functions (f_n) in $B[0,1]$ converges in the metric d.